

Yousif Shasha

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Profile

Structural Engineering student at Toronto Metropolitan University with experience in surveying, engineering laboratory testing, AutoCAD drawings, and hands-on fabrication. Brings field measurement accuracy, technical reporting, and daily team coordination experience to a construction services co-op supporting water infrastructure.

Education

Toronto Metropolitan University | Expected 2029

B.Eng. Structural Engineering | Toronto, ON

- Awards: Dean's List, Fall 2025 and Winter 2026; Top 3 Project, Geology for Engineers.
- Relevant coursework: Project Management, Surveying, Fluid Mechanics and Hydraulics, Foundation and Soils Engineering, Concrete Materials, Civil Engineering Systems, and Structural Design and Analysis.

Relevant Engineering Projects

Hydraulics Engineering | Winter 2026

EPANET, HEC-RAS, Excel

- Investigated pressurized flow, pipe losses, weirs, and sluice-gate behaviour through seven laboratory experiments.
- Built EPANET scenarios to assess the effects of pipe roughness, diameter, demand, and redundancy on water network performance.
- Compared flood profiles through the 100-year event using HEC-RAS and Excel to assess velocity and erosion risk.

Topographic Surveying | October 2025

Total Station, Excel

- Collected elevation, angle, and distance measurements with a 0.0032 m benchmark misclosure.
- Calculated traverse components, checked closure, and adjusted coordinates; validated field accuracy through direct/reverse observations and differential-leveling checks.

Structural Mechanics | Fall 2025 - Winter 2026

AutoCAD, Excel

- Produced AutoCAD drawings for aluminum, timber, and composite structural specimens; calculated stress, deflection, truss forces, and torsion by hand and in Excel.
- Compared calculations with strain-gauge and dial-gauge measurements, finding a 1.14% difference in neutral-axis location.

Geotechnical Engineering Laboratory | Winter 2026

- Classified soils using Atterberg limits and sieve analysis; evaluated compaction, permeability, and consolidation.
- Measured a 37.4-degree friction angle and assessed soil strength through direct shear and compression testing.

Leadership & Hands-on Construction

Toronto Metropolitan University Concrete Canoe Team | Sep. 2025 - Present

Construction Team Member | Toronto, ON

- Fabricated canoe components through precision cutting, sanding, taping, and assembly to meet competition requirements.
- Tested alternative materials and fabrication methods for structural performance and constructability; documented prototype results and collaborated on design improvements.

Work Experience

Groupe Adonis | July 2023 - Present

Grocery Supervisor | Mississauga, ON

- Supervise 2-6 employees, assign daily tasks, and train new hires to support organized workflows and teamwork.
- Coordinate inventory, stocking, palletizing, and placement of 10,000+ products; resolve customer and operational issues through clear communication and adaptability.

Technical Skills

Software: AutoCAD, Revit, Microsoft Excel, HEC-RAS, EPANET, ETABS (basic), MATLAB, Python, LaTeX.

Engineering: Surveying, engineering materials testing, structural analysis, data analysis, uncertainty and error analysis, technical reporting.